

UPS Battery Thermal Event - Lessons Learned

Dublin Colocation DC2 · Data Centre · Dublin, Ireland

Document ID	DC-05
Category	Power Resilience
Classification	Operational
Date	2025-09
Prepared by	European Data Centre Operations

1. What happened

On 18 June 2025, during an unusually hot spell, the single split air-conditioning unit serving UPS battery room B failed. The room temperature rose to 38°C over about three hours. The room temperature alarm was sent only by e-mail to a shared inbox and was not seen by the 24/7 network operations centre (NOC). A technician on a routine walk-round noticed the heat and found VRLA battery string 2 at 52°C with visibly swollen blocks. The string was isolated and the UPS stayed online on its remaining strings; no IT load was lost.

2. Root causes

- Battery room cooling had no redundancy: one unit and no standby.
- Critical alarms were routed to e-mail rather than to the NOC alarm console with escalation.
- Battery monitoring measured string voltage only, with no per-block temperature or internal resistance data.
- The VRLA batteries were 5.5 years old and nearing the end of their life.

3. Actions

- Per-block battery monitoring (temperature, voltage and internal resistance) on every string, with trend alarms.
- N+1 cooling for all battery rooms, and high-temperature alarms routed to the NOC with automatic escalation to the duty manager.
- VRLA strings replaced with lithium-ion batteries with an integrated battery management system and thermal-runaway detection.
- Quarterly thermal imaging of battery rooms and switchgear.
- Integrated systems test of the UPS, generators and cooling twice a year.

4. Lessons learned

- An alarm that goes to e-mail is not an alarm. Every critical alarm needs a monitored console and an escalation path.
- Single points of failure hide in support spaces: battery rooms, switch rooms and pump rooms deserve the same redundancy review as the data halls.
- Monitor what fails first: for batteries that is temperature and internal resistance, not string voltage.
- Test like you mean it: integrated tests under real load reveal interactions that component tests miss.